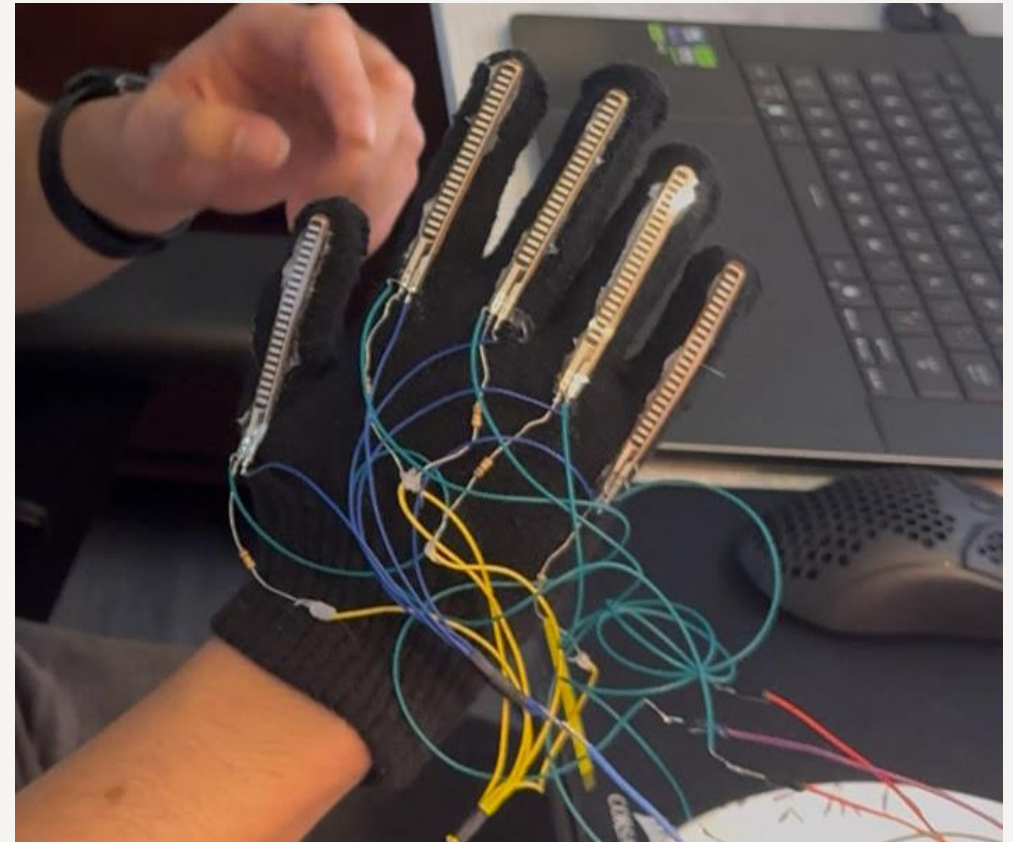


Sign-Language-Glove

AYMAN EL-HINDI AND LE QUOC VIET PHAM

Introduction of the Project

- A wearable Arduino device
- Translates hand gestures into text
- Designed to help people communicate sign-language



Project Features

01

Input: 5 flex sensors for each fingers

02

Decode: determine which fingers are flexed

03

Output: display the corresponding message

Project Features

CODE SAMPLE 1 (finger assignment):

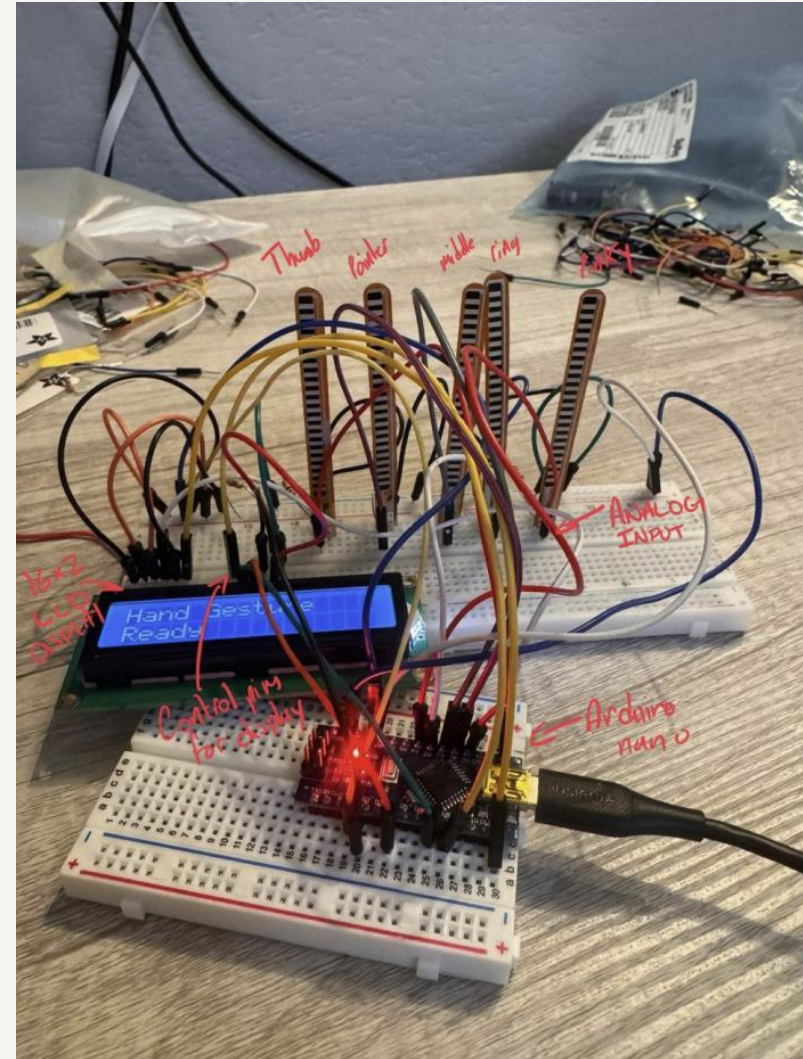
```
bool t = thumb > flexThreshold;  
  
bool p = pointer > flexThreshold;  
  
bool m = middle > flexThreshold;  
  
bool r = ring > flexThreshold;  
  
bool k = pinky > flexThreshold;
```

CODE SAMPLE 2 (condition for "Good Job!"
Output):

```
    else if (!t && p && m && r && k) {  
  
        lcd_1.clear();  
  
        lcd_1.setCursor(0, 0); // Changed from (0, 2)  
to (0, 0)  
  
        lcd_1.print("Good Job!");  
  
        Serial.println("gg");
```

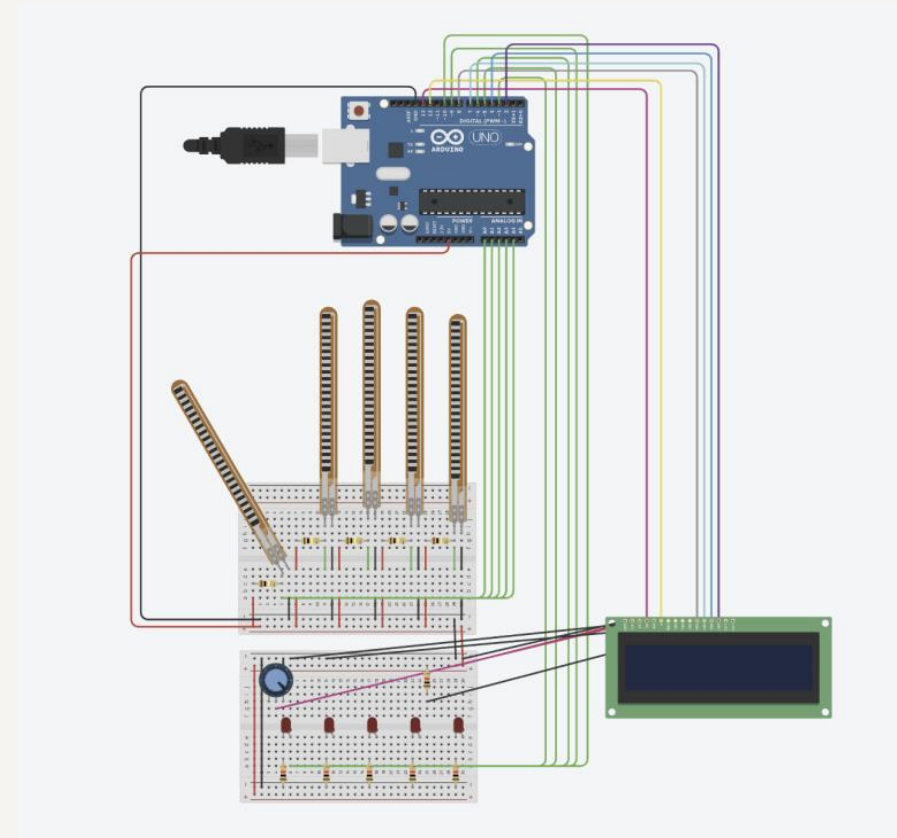
Project Features

Output: text on LCD Screen

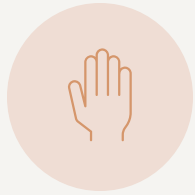


Challenges Encountered + How We Overcame Them

- Calibrating the sensors and testing code without physical hardware
==> used Tinkercad simulations to build and test the circuit and code collaboratively online
- The value on Tinkercad didn't match the actual value
==> used the Serial Monitor to fine-tune the sensor thresholds on the actual hardware



Features We Would Like To Add



Transition from a messy breadboard prototype to a clean, wearable glove with audio.



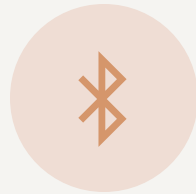
Solder the components, use a ribbon cable for the LCD, and add a battery



Expand our current code matching with ASL



Add sound clips from a speaker for each gesture.



Text output on phone screen through Wifi/Bluetooth



Adding **hand motion** (ex: waving = hi)